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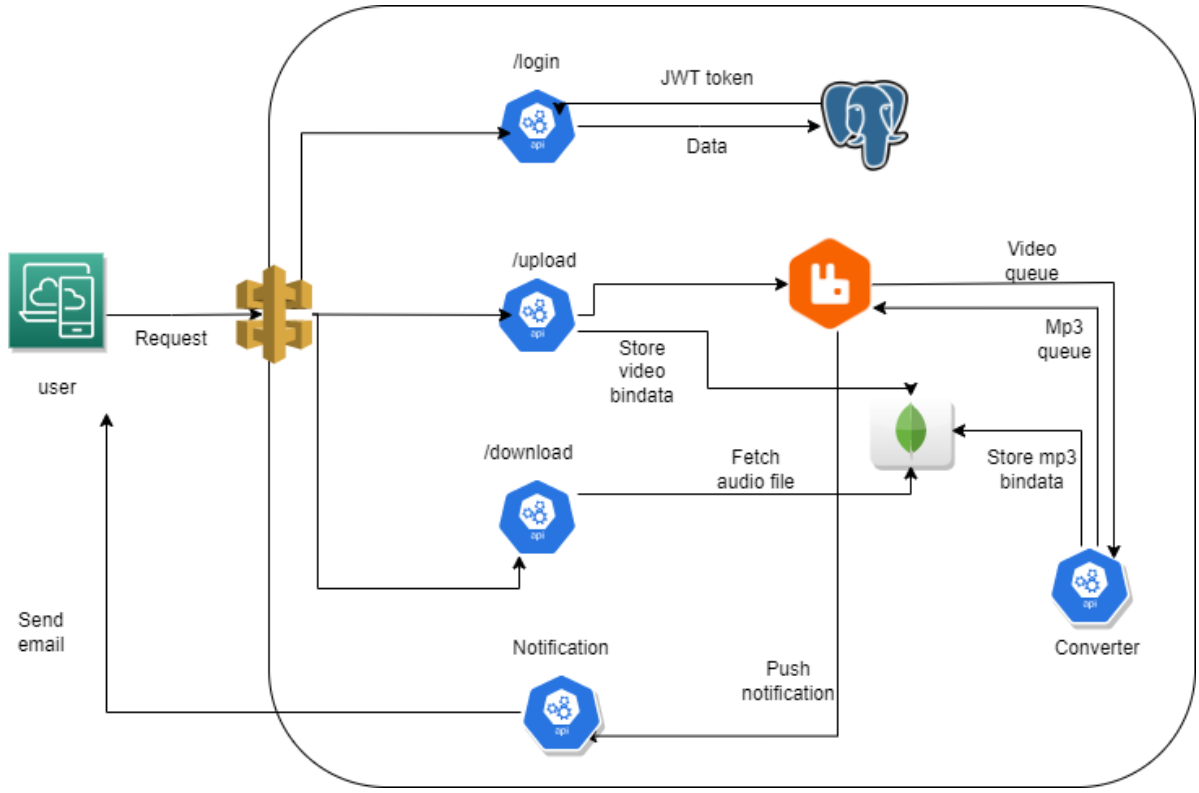
End-to-End Deployment of a Python-Based Microservice Application on AWS EKS: A Comprehensive Guide

GITHUB REPO - https://github.com/Ahmedgit7/video-audio-microservices-python-app

In this project we will deploy a python microservices application which is going to convert video into audio and deploy it on AWS Cloud using tools like Kubernetes, Helm, Docker, rabbit MQ and also databases including postgresql and Mongo DB.

Architecture:



In this application, we have four different microservices: the Auth Service, Converter Service, Gateway Service, and Notification Service. Additionally, we use Helm charts to deploy RabbitMQ, MongoDB, and PostgreSQL.

The application works as follows:

- The user interacts with the API Gateway, which has three endpoints: login, upload, and download.
- First, the user logs in. The Auth Service checks the credentials against the PostgreSQL database. If the credentials are correct, a JWT token is generated, which allows the user to upload a video.

- When a user uploads a video via the upload endpoint, the video is stored in RabbitMQ. The video is then transferred to the Converter Service, which converts the video into audio and stores the converted data in MongoDB (deployed via Helm charts).
- Once the video is converted to audio, the Notification Service sends a notification to the user's email address.
- Finally, the user can download the converted audio via the download endpoint, accessing the file through the Gateway Service.

INFRASTRUCTURE HAS BEEN BUILT USING TERRAFORM, THE CODES ARE ADDED IN THE GITHUB REPO.

Prerequisites

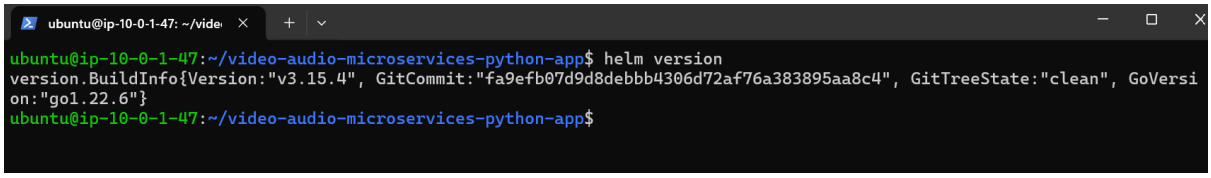
1.Install Helm:

Commands:

$ curl -fsSL -o get_helm.sh
https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3

$ chmod 700 get_helm.sh

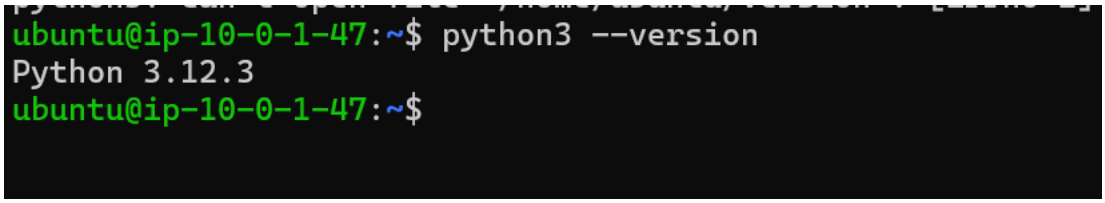
$ ./get_helm.sh



**

2.Python:

Already installed



**

3. AWS CLI:

Commands:

“curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"

unzip awscliv2.zip

sudo ./aws/install"

After installing awscli do aws configure

```
ubuntu@ip-10-0-1-47:~$ aws configure
AWS Access Key ID [****************BB2Y]:
AWS Secret Access Key [****************LBvK]:
Default region name [ap-south-1]:
Default output format [None]:
ubuntu@ip-10-0-1-47:~$
```

**

4. Install kubectl:

Commands:

"curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl

chmod +x kubectl

sudo mv kubectl /usr/local/bin"

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app$ kubectl version
Client Version: v1.31.0
Kustomize Version: v5.4.2
The connection to the server localhost:8080 was refused - did you specify the right host or port?
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app$
```

**

5. Databases:

Mongodb: if we don't have mongodb installed in our machine then we need to install mongodb in it.

Although we will be installing mongodb in kubernetes cluster using helm charts, the mongosh command isn't automatically available on your local machine because it is a client tool that needs to be installed separately from the MongoDB server. The Helm chart installation sets up MongoDB on your Kubernetes cluster, but you still need mongosh locally to interact with it.

So for that we need to install mongodb in our machine using this commands: AS I AM USING UBUNTU:

"curl -fsSL https://www.mongodb.org/static/pgp/server-4.4.asc | sudo apt-key add –

apt-key list

echo "deb [arch=amd64,arm64] https://repo.mongodb.org/apt/ubuntu bionic/mongodb-org/4.4 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-4.4.list

sudo apt update

sudo apt install mongodb-org"

Postgres Installation:

"sudo apt install postgresql"

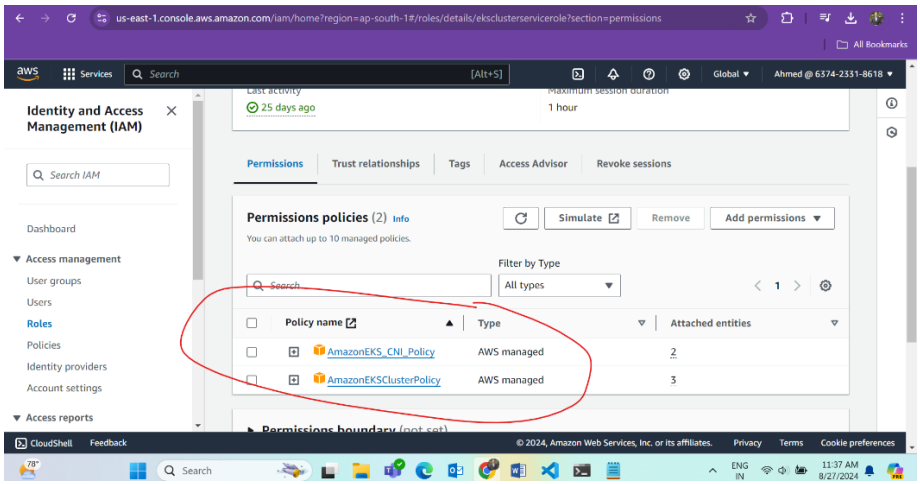
**

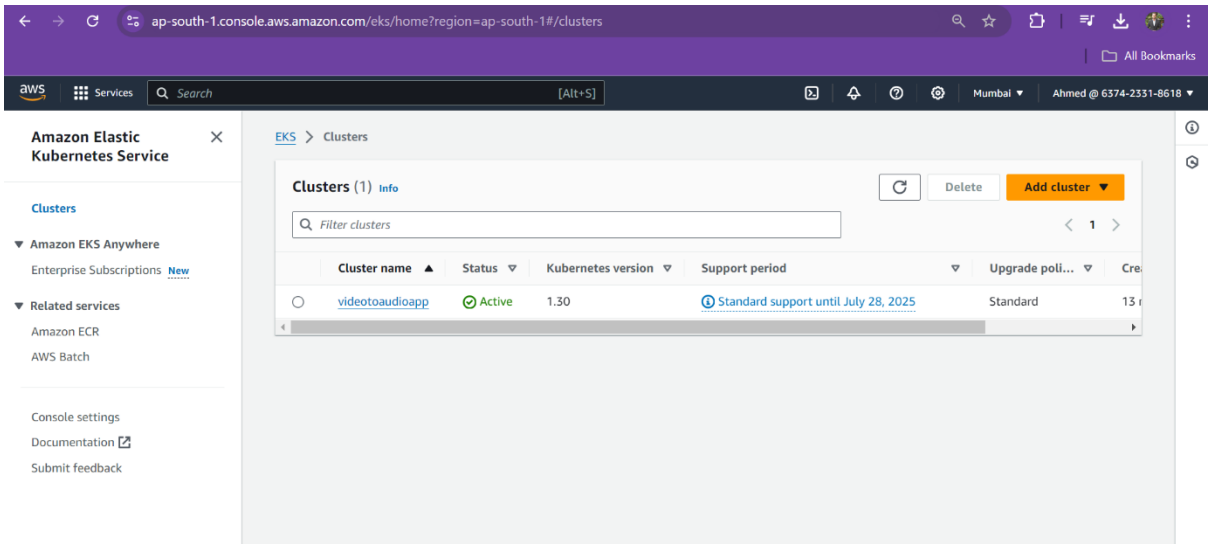
STEP 1: Clone the Repository to our machine:

"git clone https://github.com/Ahmedgit7/video-audio-microservices-python-app.git"

STEP 2: Create a Kubernetes Cluster.

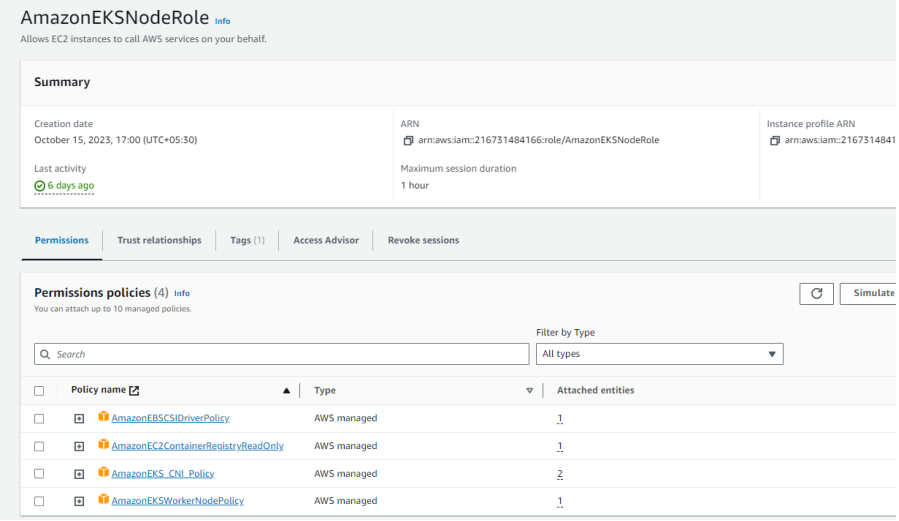
Create a Role for kubernetes cluster which include these policies attached.



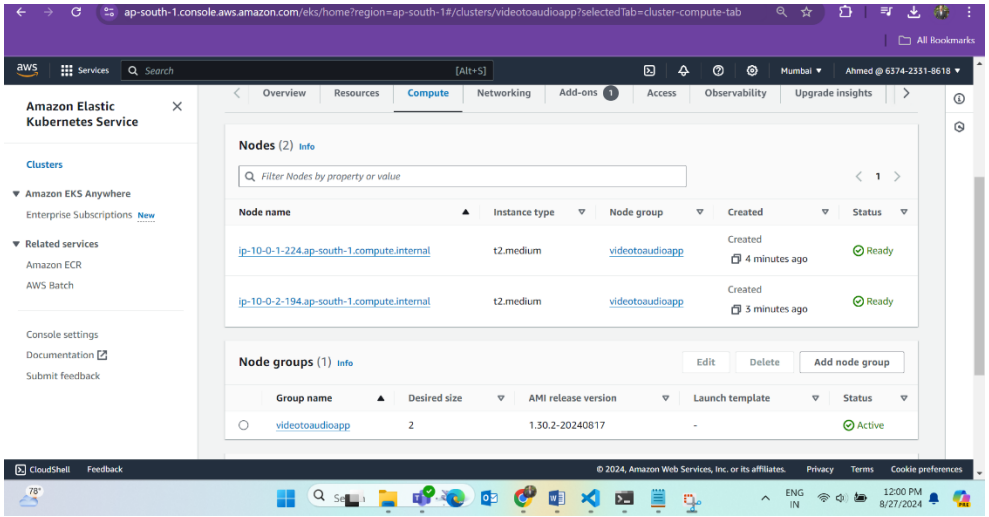


Create Nodegroup:

Create an IAM Role for this Nodegroup which should contain these below policies:

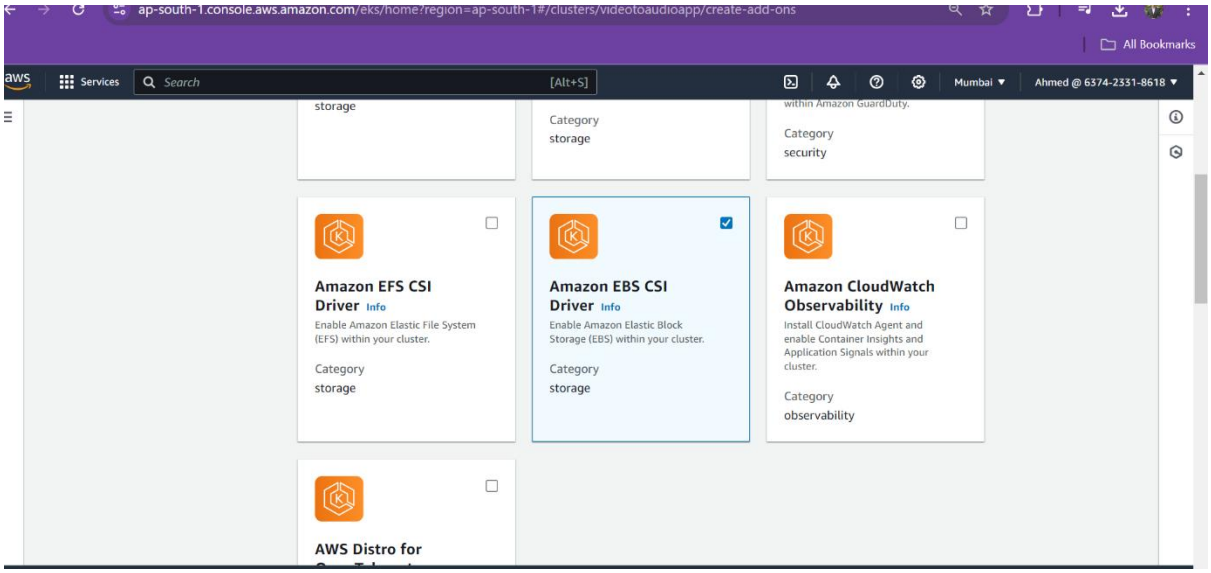


Nodegroup has been Created:



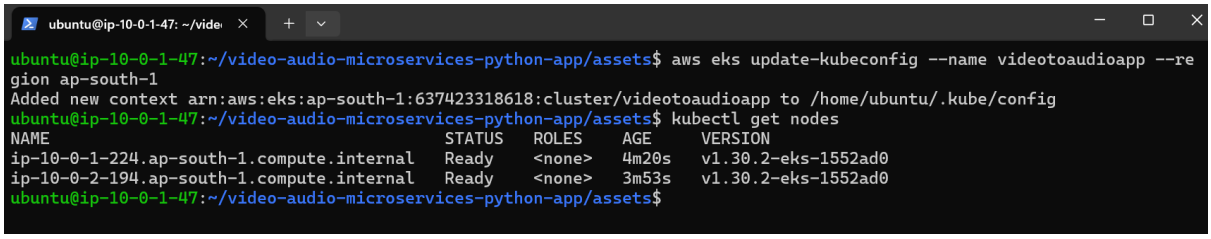
Enable EBS CSI Addon

Enable addon ebs csi this is for enabling pvcs once cluster is created



Connected Kubernetes Cluster with cli:

“aws eks update-kubeconfig --name <cluster_name> --region <aws_region>”

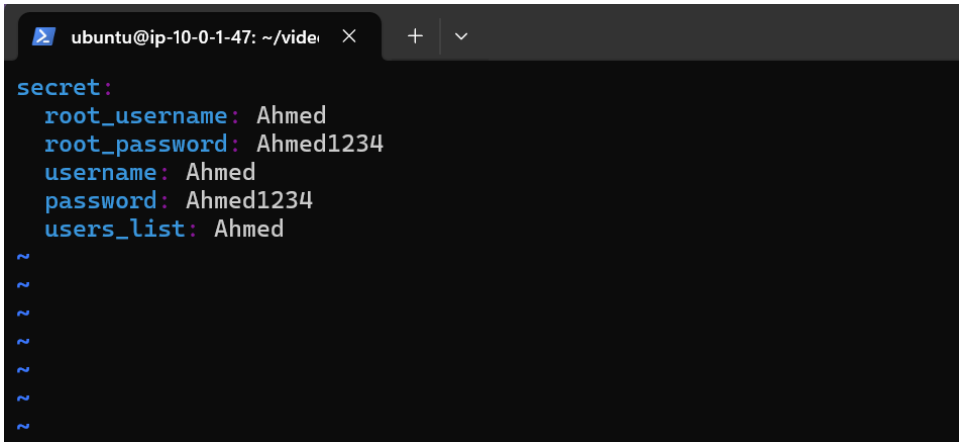


```
ubuntu@ip-10-0-1-47: ~/vide
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/assets$ aws eks update-kubeconfig --name videotoaudioapp --re
gion ap-south-1
Added new context arn:aws:eks:ap-south-1:637423318618:cluster/videotoaudioapp to /home/ubuntu/.kube/config
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/assets$ kubectl get nodes
NAME                                          STATUS   ROLES    AGE     VERSION
ip-10-0-1-224.ap-south-1.compute.internal     Ready    <none>   4m20s   v1.30.2-eks-1552ad0
ip-10-0-2-194.ap-south-1.compute.internal     Ready    <none>   3m53s   v1.30.2-eks-1552ad0
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/assets$
```

Installation of MongoDB, Postgres, RabbitMQ:

1.MongoDB

To install MongoDB, set the database username and password in values.yaml,



```
ubuntu@ip-10-0-1-47: ~/vide
secret:
  root_username: Ahmed
  root_password: Ahmed1234
  username: Ahmed
  password: Ahmed1234
  users_list: Ahmed
~
~
~
~
~
~
~
```

Then navigate to the MongoDB Helm chart folder and run:

“cd Helm_charts/MongoDB

helm install mongo .”

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/MongoDB$ helm install mongo .
NAME: mongo
LAST DEPLOYED: Tue Aug 27 06:41:46 2024
NAMESPACE: default
STATUS: deployed
REVISION: 1
TEST SUITE: None
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/MongoDB$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
mongodb-0   1/1     Running   0          23s
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/MongoDB$ kubectl get all
NAME            READY   STATUS    RESTARTS   AGE
pod/mongodb-0   1/1     Running   0          56s

NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)           AGE
service/kubernetes   ClusterIP   172.20.0.1      <none>        443/TCP           25m
service/mongodb      NodePort    172.20.98.153   <none>        27017:30005/TCP   56s

NAME                       READY   AGE
statefulset.apps/mongodb   1/1     56s
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/MongoDB$
```

Persistent volume is also created for the mongodb

```
ubuntu@ip-10-0-1-47: ~

ubuntu@ip-10-0-1-47:~$ kubectl get pvc
NAME          STATUS   VOLUME       CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
mongodb-pvc   Bound    mongodb-pv   10Gi       RWO            manual         <unset>                 2m50s
ubuntu@ip-10-0-1-47:~$
```

2. Postgres

PostgreSQL

Set the database username and password in values.yaml. Install PostgreSQL from the PostgreSQL Helm chart folder and initialize it with the queries in init.sql.

```
ubuntu@ip-10-0-1-47: ~/vide

service:
  name: db
  port: 5432
  nodeport: 30003

container:
  image: postgres
  env:
    user: Ahmed
    password: Ahmed2023
    db: authdb
~
~
~
```

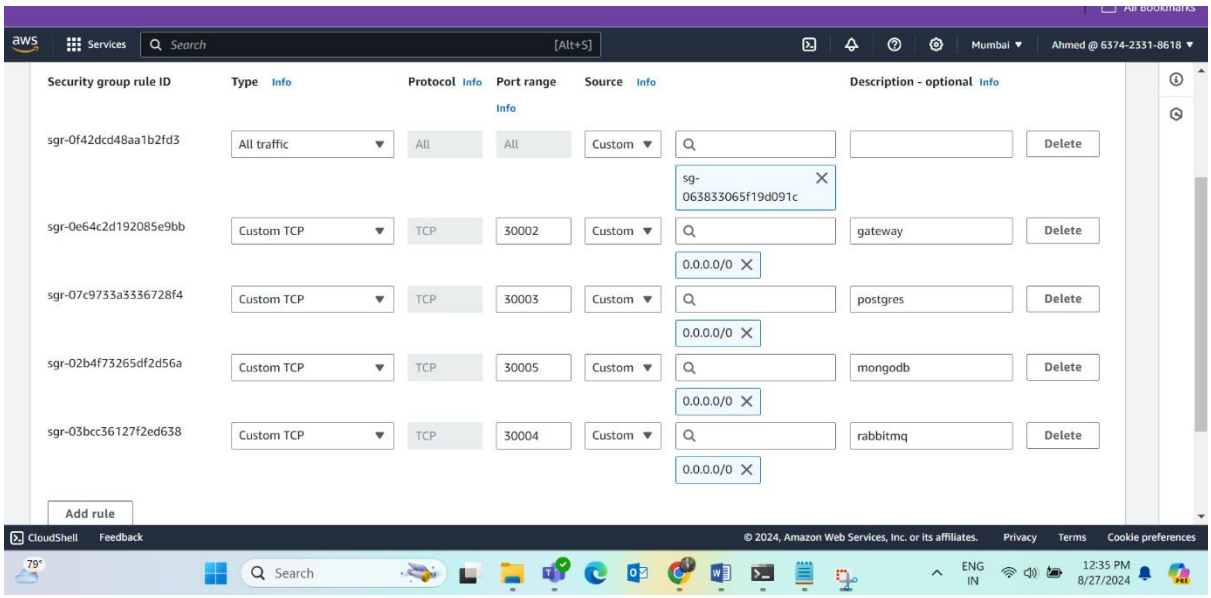
Go to init.sql file and give the email id to which you need to get the notification:

```
ubuntu@ip-10-0-1-47: ~/vide

CREATE TABLE auth_user (
    id integer GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    email VARCHAR (255) NOT NULL,
    password VARCHAR (255) NOT NULL
);

--Add Username and Password for Admin User
-- INSERT INTO auth_user (email, password) VALUES ('thomasfookins007helby@gmail.com', '123456');
INSERT INTO auth_user (email, password) VALUES ('ahmedhshaikh786@gmail.com', '123456');
~
~
~
~
```

Postgres running:

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/Postgres$ helm install postgres .
NAME: postgres
LAST DEPLOYED: Tue Aug 27 06:57:20 2024
NAMESPACE: default
STATUS: deployed
REVISION: 1
TEST SUITE: None
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/Postgres$ kubectl get all
NAME                                    READY   STATUS    RESTARTS   AGE
pod/mongodb-0                           1/1     Running   0          15m
pod/postgres-deploy-7784554477-9rbjk    1/1     Running   0          14s

NAME                 TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)           AGE
service/db           NodePort    172.20.189.206   <none>        5432:30003/TCP    14s
service/kubernetes   ClusterIP   172.20.0.1       <none>        443/TCP           40m
service/mongodb      NodePort    172.20.98.153    <none>        27017:30005/TCP   15m

NAME                              READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/postgres-deploy   1/1     1            1           14s

NAME                                         DESIRED   CURRENT   READY   AGE
replicaset.apps/postgres-deploy-7784554477   1         1         1       14s

NAME                      READY   AGE
statefulset.apps/mongodb  1/1     15m
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/Postgres$ kubectl get pods
NAME                                    READY   STATUS    RESTARTS   AGE
mongodb-0                               1/1     Running   0          15m
postgres-deploy-7784554477-9rbjk        1/1     Running   0          19s
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/Postgres$
```

Update the security group of nodes:



Once the Installation of Postgres and Mongodb has been completed, we will go and connect to Mongodb and postgresql.

“mongosh mongodb://<username>:<pwd>@<nodeip>:30005/mp3s?authSource=admin”

Username and password of – values.yml

Connected to Mongodb

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/Postgres$ mongosh mongodb://Ahmed:Ahmed1234@3.111
.169.32:30005/mp3s?authSource=admin
Current Mongosh Log ID: 66cd7d4223546c5cbc5e739b
Connecting to:          mongodb://<credentials>@3.111.169.32:30005/mp3s?authSource=admin&directConnection=true&appName=m
ongosh+2.3.0
Using MongoDB:          4.0.8
Using Mongosh:          2.3.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/


To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.co
m/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

mp3s>
```

Basically there is nothing in our mongodb db, So its purpose is to store the videos and audio which gets converted.

(Note If you have changed the username and password of mongodb in values.yml then, make sure to change them in configmap of all the manifests file)

Now we will exit from mongodb and get connect to postgresql.

“psql 'postgres://<username>:<pwd>@<nodeip>:30003/authdb'”

Username and password from values.yml

Successfully connected

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/Postgres$ psql 'postgres://Ahmed:Ahmed2023@3.111.
169.32:30003/authdb'
psql (16.4 (Ubuntu 16.4-0ubuntu0.24.04.2))
Type "help" for help.

authdb=#
```

Now here we need to create a table called auth_user and insert user credentials into it.

CREATE TABLE auth_user (

id integer GENERATED ALWAYS AS IDENTITY PRIMARY KEY,

email VARCHAR (255) NOT NULL,

password VARCHAR (255) NOT NULL

);

--Add Username and Password for Admin User

-- INSERT INTO auth_user (email, password) VALUES ('thomasfookins007helby@gmail.com', '123456');

INSERT INTO auth_user (email, password) VALUES ('ahmedhshaikh786@gmail.com', '123456');

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/Postgres$ psql 'postgres://Ahmed:Ahmed2023@3.111.
169.32:30003/authdb'
psql (16.4 (Ubuntu 16.4-0ubuntu0.24.04.2))
Type "help" for help.

authdb=# CREATE TABLE auth_user (
    id integer GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    email VARCHAR (255) NOT NULL,
    password VARCHAR (255) NOT NULL
);

--Add Username and Password for Admin User
-- INSERT INTO auth_user (email, password) VALUES ('thomasfookins007helby@gmail.com', '123456');
INSERT INTO auth_user (email, password) VALUES ('ahmedhshaikh786@gmail.com', '123456');
CREATE TABLE
INSERT 0 1
```

```
authdb=# \d
              List of relations
 Schema |       Name       |   Type   | Owner
--------+------------------+----------+-------
 public | auth_user        | table    | Ahmed
 public | auth_user_id_seq | sequence | Ahmed
(2 rows)

authdb=# SELECT * FROM auth_user;
 id |           email           | password
----+---------------------------+----------
  1 | ahmedhshaikh786@gmail.com | 123456
(1 row)
```

Install RabbitMQ:

Go to the Rabbitmq folder and run the following command to install rabbitmq using helm:

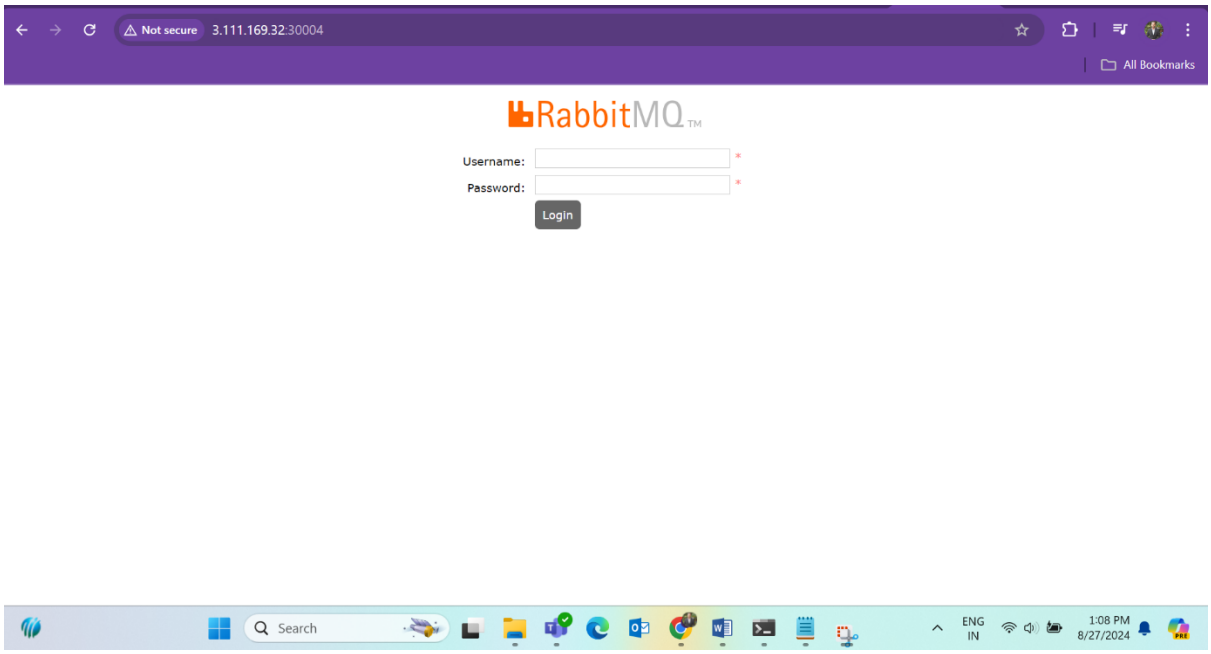
“helm install rabbitmq .”

You can see the rabbit mq has been installed successfully and pod is running.

```
ubuntu@ip-10-0-1-47: ~/vide
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/RabbitMQ/templates$ cd ..
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/RabbitMQ$ helm install rabbitmq .
NAME: rabbitmq
LAST DEPLOYED: Tue Aug 27 07:36:56 2024
NAMESPACE: default
STATUS: deployed
REVISION: 1
TEST SUITE: None
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/RabbitMQ$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
mongodb-0                           1/1     Running             0          55m
postgres-deploy-7784554477-9rbjk    1/1     Running             0          39m
rabbitmq-0                          0/1     ContainerCreating   0          18s
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/RabbitMQ$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
mongodb-0                           1/1     Running             0          55m
postgres-deploy-7784554477-9rbjk    1/1     Running             0          39m
rabbitmq-0                          0/1     ContainerCreating   0          21s
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/RabbitMQ$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mongodb-0                           1/1     Running   0          55m
postgres-deploy-7784554477-9rbjk    1/1     Running   0          39m
rabbitmq-0                          1/1     Running   0          23s
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/Helm_charts/RabbitMQ$
```

Now we will go access the rabbitmq dashboard on browser.

Access it using - <nodeip>:30004

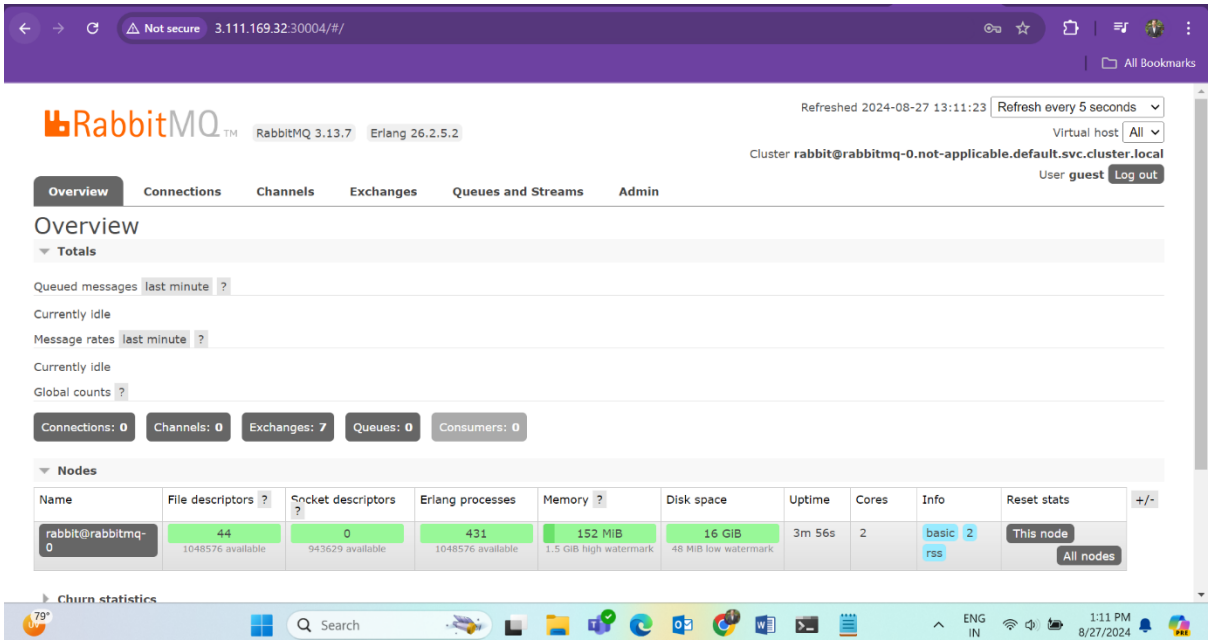


Username – guest

Password – guest

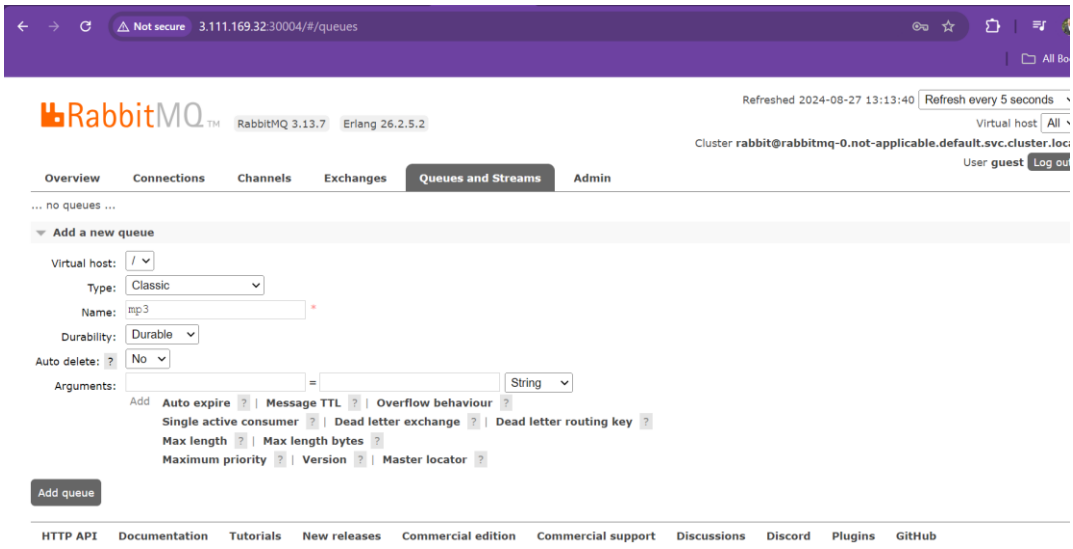
Once we log in into the rabbit mq dashboard we will go and create a queues.

In RabbitMQ, a queue is a buffer that stores messages. Queues are the main components where messages are sent to, stored, and consumed from. They are integral to RabbitMQ's messaging system, which allows communication between different parts of a system or between different applications.

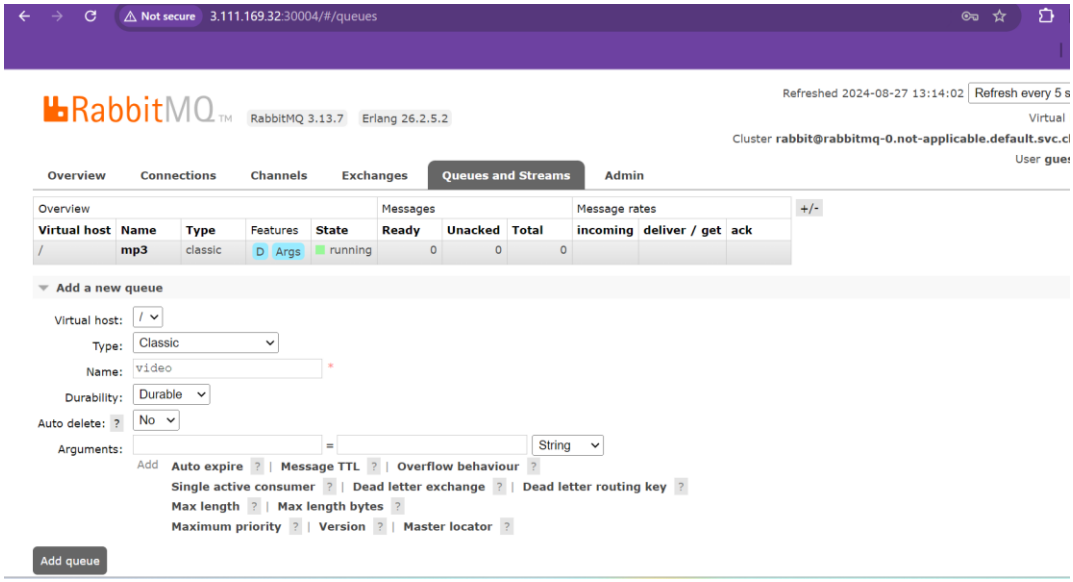


Go to Queues One for MP3 and One for Video.

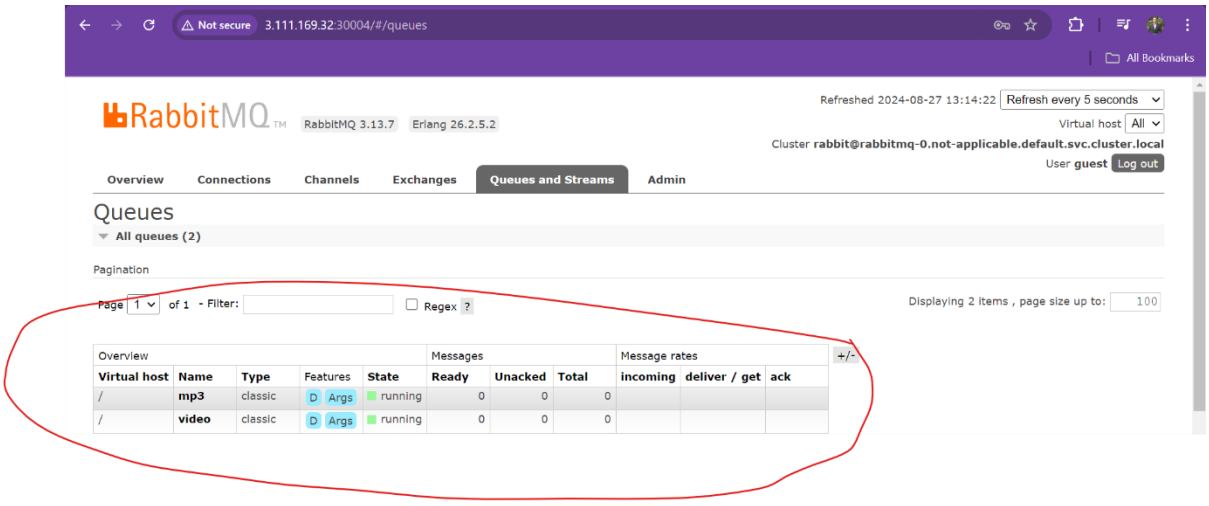
1.MP3 Queue



2. Video Queue



Queues Created



Once we have installed Mongodb, Postgresql and RabbitMQ, the next step is to install our microservices individualy to create a application and validate if we are able to convert video to audio or not.

1.auth-service

Go to Project Folder>src>auth_service

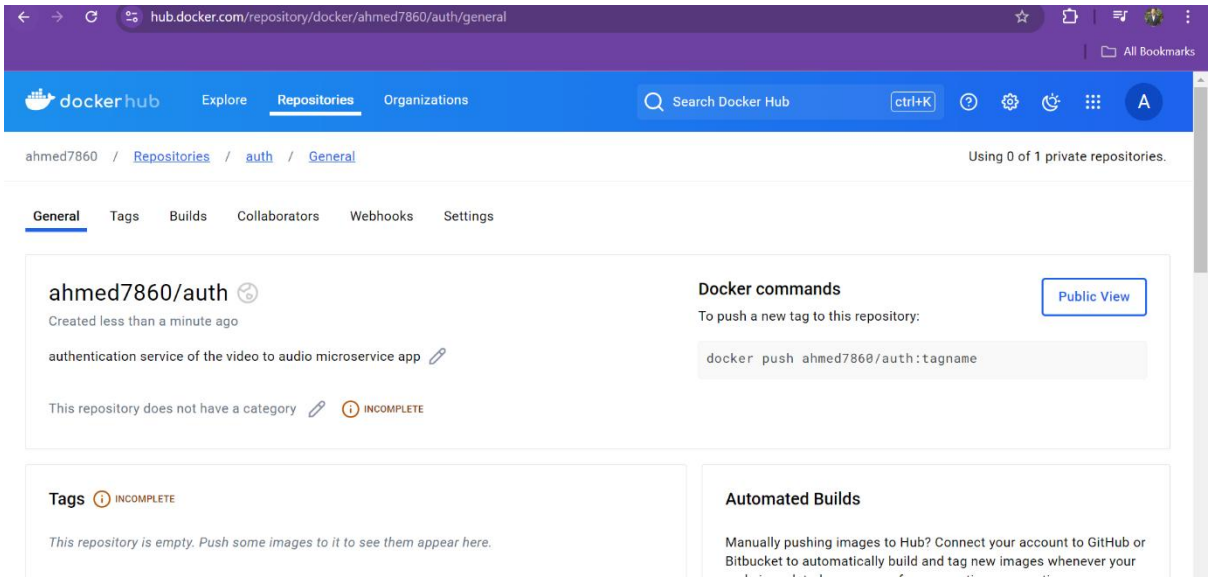
And built the image for this service.

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/auth-service$ sudo docker build -t auth .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  11.26kB
Step 1/8 : FROM python:3.10-slim-bullseye
3.10-slim-bullseye: Pulling from library/python
82aabceedc2f: Pull complete
92566c25ddaa: Pull complete
99e8ea822ad1: Pull complete
c74e2b8db7e7: Pull complete
b05e7fcaa1f6: Pull complete
Digest: sha256:cade10e6feea11d3ee1eb569d4da345e4d7085574f0d2112f5f7a563427ddb83
Status: Downloaded newer image for python:3.10-slim-bullseye
 ---> 0818d4906b16
Step 2/8 : RUN apt-get update && apt-get install -y --no-install-recommends --no-install-suggests build-essential libpq-
dev python3-dev && pip install --no-cache-dir --upgrade pip
```

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/auth-service$ sudo docker images
REPOSITORY   TAG                 IMAGE ID       CREATED              SIZE
auth         latest              027ab4a34a04   About a minute ago   503MB
python       3.10-slim-bullseye  0818d4906b16   4 weeks ago          125MB
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/auth-service$
```

After creating an image we need to push the image to dockerhub, for that first we need to login into our dockerhub. And create one repository.

So for this I have created auth repo.




```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/auth-service$ sudo docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head
 over to https://hub.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and
 is required for organizations using SSO. Learn more at https://docs.docker.com/go/access-tokens/

Username: ahmed7860
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
```

After creating repo in docker hub go and push the docker image which has been built before to docker hub repo.

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/auth-service$ sudo docker push ahmed7860/auth
Using default tag: latest
The push refers to repository [docker.io/ahmed7860/auth]
f87ad4dae50b: Pushed
cfb574673340: Pushed
e6f200459f72: Pushed
c51dbbed1111: Pushed
0389280d6ad0: Pushed
acc5a7b3e8d6: Mounted from library/python
195a4622ada2: Mounted from library/python
c0f8c3cb14e1: Mounted from library/python
0a49aba3fd52: Mounted from library/python
bf63823f8c99: Mounted from library/python
latest: digest: sha256:9251afc2174f12491ba9e8d36d0967222179fc02a3b272931dc5668a3fa2e5a3 size: 2416
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/auth-service$
```

2. Converter

Build the image for this using above steps:

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/converter-service$ sudo docker build -t converter .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  15.87kB
Step 1/7 : FROM python:3.10-slim-bullseye
 ---> 0818d4906b16
Step 2/7 : RUN apt-get update && apt-get install -y --no-install-recommends --no-install-suggests build-essential libpq-dev python3-d
ev ffmpeg && pip install --no-cache-dir --upgrade pip
 ---> Running in 8e79de0116fa
Get:1 http://deb.debian.org/debian bullseye InRelease [116 kB]
Get:2 http://deb.debian.org/debian-security bullseye-security InRelease [48.4 kB]
Get:3 http://deb.debian.org/debian bullseye-updates InRelease [44.1 kB]
Get:4 http://deb.debian.org/debian bullseye/main amd64 Packages [8067 kB]
Get:5 http://deb.debian.org/debian-security bullseye-security/main amd64 Packages [280 kB]
Get:6 http://deb.debian.org/debian bullseye-updates/main amd64 Packages [18.8 kB]
Fetched 8574 kB in 2s (5330 kB/s)
```

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/converter-service$ sudo docker images
REPOSITORY        TAG                 IMAGE ID       CREATED          SIZE
converter         latest              45387fcc1c34   5 seconds ago    1.03GB
ahmed7860/auth    latest              aea68383cf4c   5 minutes ago    503MB
auth              latest              aea68383cf4c   5 minutes ago    503MB
ahmed7860/auth    <none>              027ab4a34a04   18 minutes ago   503MB
python            3.10-slim-bullseye  0818d4906b16   4 weeks ago      125MB
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/converter-service$ sudo docker tag converter:latest ahmed7860/converte
r:latest
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/converter-service$ sudo docker push ahmed7860/converter:latest
The push refers to repository [docker.io/ahmed7860/converter]
32623acbcc72: Pushed
38fa4dafa15a: Pushed
b46058949fd9: Pushed
7ddb0e4eed5e: Pushed
3772a664ab80: Pushed
acc5a7b3e8d6: Mounted from ahmed7860/auth
```

3. gateway

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/gateway-service$ sudo docker build -t gateway .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  18.43kB
Step 1/8 : FROM python:3.10-slim-bullseye
 ---> 0818d4906b16
Step 2/8 : RUN apt-get update && apt-get install -y --no-install-recommends --no-install-suggests build-essential libpq-dev python3-d
ev && pip install --no-cache-dir --upgrade pip
 ---> Using cache
 ---> e7479397ac17
Step 3/8 : WORKDIR /app
 ---> Using cache
 ---> 1651d39afa61
Step 4/8 : COPY ./requirements.txt /app
```

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/gateway-service$ sudo docker tag gateway:latest ahmed7860/gateway
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/gateway-service$ sudo docker push ahmed7860/gateway
Using default tag: latest
The push refers to repository [docker.io/ahmed7860/gateway]
2fc6ee363076: Pushed
361d428211c4: Pushed
19a948576158: Pushed
c51dbbed1111: Mounted from ahmed7860/auth
0389280d6ad0: Mounted from ahmed7860/auth
acc5a7b3e8d6: Mounted from ahmed7860/converter
195a4622ada2: Mounted from ahmed7860/converter
c0f8c3cb14e1: Mounted from ahmed7860/converter
0a49aba3fd52: Mounted from ahmed7860/converter
bf63823f8c99: Mounted from ahmed7860/converter
latest: digest: sha256:a06b9f0213b113f62b94d7c26faf8b4feadaaf52ff23d6cee02e448c7fcac03c size: 2416
```

4. notification

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/notification-service$ sudo docker build -t notification .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  11.26kB
Step 1/7 : FROM python:3.10-slim-bullseye
 ---> 0818d4906b16
Step 2/7 : RUN apt-get update && apt-get install -y --no-install-recommends --no-install-suggests build-essential libpq-dev python3-d
ev ffmpeg && pip install --no-cache-dir --upgrade pip
 ---> Using cache
 ---> 472afc6fb0b7
Step 3/7 : WORKDIR /app
```

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/notification-service$ sudo docker images
REPOSITORY            TAG                 IMAGE ID       CREATED          SIZE
notification          latest              f25425959ddc   22 seconds ago   858MB
ahmed7860/gateway     latest              c5bc0f48f5bd   4 minutes ago    491MB
gateway               latest              c5bc0f48f5bd   4 minutes ago    491MB
ahmed7860/converter   latest              45387fcc1c34   7 minutes ago    1.03GB
converter             latest              45387fcc1c34   7 minutes ago    1.03GB
ahmed7860/auth        latest              aea68383cf4c   13 minutes ago   503MB
auth                  latest              aea68383cf4c   13 minutes ago   503MB
ahmed7860/auth        <none>              027ab4a34a04   26 minutes ago   503MB
python                3.10-slim-bullseye  0818d4906b16   4 weeks ago      125MB
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/notification-service$ sudo docker tag notification:latest ahmed7860/no
tification
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/notification-service$ sudo docker push ahmed7860/notification
Using default tag: latest
The push refers to repository [docker.io/ahmed7860/notification]
9ec27deba5ab: Pushed
51afe81a216b: Pushed
789ba2221ade: Pushed
7ddb0e4eed5e: Mounted from ahmed7860/converter
3772a664ab80: Mounted from ahmed7860/converter
acc5a7b3e8d6: Mounted from ahmed7860/gateway
195a4622ada2: Mounted from ahmed7860/gateway
c0f8c3cb14e1: Mounted from ahmed7860/gateway
0a49aba3fd52: Mounted from ahmed7860/gateway
```

Apply the manifest file for each microservice:

1.auth manifest file applied.

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/auth-service$ cd manifest/
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/auth-service/manifest$ ls
configmap.yaml  deployment.yaml  secret.yaml  service.yaml
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/auth-service/manifest$ kubectl apply -f .
configmap/auth-configmap created
deployment.apps/auth created
secret/auth-secret created
service/auth created
```

Auth Pods are running

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src$ kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
auth-5db58548fd-m8zvw         1/1     Running   0          9m58s
auth-5db58548fd-vn4db         1/1     Running   0          9m58s
```

2. gateway manifest file applied

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/gateway-service$ cd manifest/
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/gateway-service/manifest$ ls
configmap.yaml  gateway-deploy.yaml  secret.yaml  service.yaml
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/gateway-service/manifest$ kubectl apply -f .
configmap/gateway-configmap created
deployment.apps/gateway created
secret/gateway-secret created
service/gateway created
```

Gateway pods are running

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/gateway-service/manifest$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
auth-5db58548fd-m8zvw               1/1     Running   0          11m
auth-5db58548fd-vn4db               1/1     Running   0          11m
gateway-b5ddf9574-lbb5v             1/1     Running   0          11s
gateway-b5ddf9574-rxbkj             1/1     Running   0          11s
```

3. converter-service file is applied

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/converter-service/manifest$ ls
configmap.yaml  converter-deploy.yaml  secret.yaml
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/converter-service/manifest$ kubectl apply -f .
configmap/converter-configmap created
deployment.apps/converter created
secret/converter-secret created
```

Converter pods are running

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/converter-service/manifest$ kubectl get po
NAME                                READY   STATUS    RESTARTS   AGE
auth-5db58548fd-m8zvw               1/1     Running   0          13m
auth-5db58548fd-vn4db               1/1     Running   0          13m
converter-74dc46b684-fnsp5          1/1     Running   0          28s
converter-74dc46b684-k4lkl          1/1     Running   0          28s
converter-74dc46b684-nv9t7          1/1     Running   0          28s
converter-74dc46b684-t4dzt          1/1     Running   0          28s
```

4. notification manifest file is applied

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/notification-service/manifest$ kubectl apply -f .
configmap/notification-configmap created
deployment.apps/notification created
secret/notification-secret created
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/notification-service/manifest$ kubectl get pods
```

Notification pods are running.

```
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/notification-service/manifest$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
auth-5db58548fd-m8zvw               1/1     Running   0          14m
auth-5db58548fd-vn4db               1/1     Running   0          14m
converter-74dc46b684-fnsp5          1/1     Running   0          99s
converter-74dc46b684-k4lkl          1/1     Running   0          99s
converter-74dc46b684-nv9t7          1/1     Running   0          99s
converter-74dc46b684-t4dzt          1/1     Running   0          99s
gateway-b5ddf9574-lbb5v             1/1     Running   0          3m10s
gateway-b5ddf9574-rxbkj             1/1     Running   0          3m10s
mongodb-0                           1/1     Running   0          3h5m
notification-75c5f96c59-bxqfc       1/1     Running   0          21s
notification-75c5f96c59-wbqtn       1/1     Running   0          21s
postgres-deploy-7784554477-9rbjk    1/1     Running   0          169m
rabbitmq-0                          1/1     Running   0          130m
ubuntu@ip-10-0-1-47:~/video-audio-microservices-python-app/src/notification-service/manifest$
```

Once our Pods are running Not let us see How we can convert the video into audio part.

For that go to cd projectfolder/assets folder where your video is stored.

```
ubuntu@ip-10-5-1-157:~/microservices-python-app/assets$ ls
Rockstar.mp4  video.mp4
```

And then run this below login endpoint:

“curl -X POST http://nodeIP:30002/login -u <email>:<password>”

```
ubuntu@ip-10-5-1-157:~/microservices-python-app/assets$ curl -X POST http://13.235.76.100:30002/login -u ahmedhshaikh786@gmail.com:123456
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VybmFtZSI6ImFobWVkaHNoYWlraDc4NkBnbWFpbC5jb20iLCJleHAiOjE3MjUwMTY3NDYsImlhdCI6MTcyNDkzMDM0NiwiYWRtaW4iOnRy
dWV9.hDIZlv36qo0y5ET8waCT47sNukRzNjDgUr8x4jJGeS0ubuntu@ip-10-5-1-157:~/microservices-python-app/assets$ ^C
```

As you can see we have successfully logged in with the gmail and password which we have stored in the postgresql. So when user logged in using those creds postgresql checked it and provided us the jwt token which we will be using it to upload the video to convert it to audio.

To Upload Run this below Upload Endpoint:

"curl -X POST -F 'file=@./video.mp4' -H 'Authorization: Bearer <JWT Token>' http://nodeIP:30002/upload"

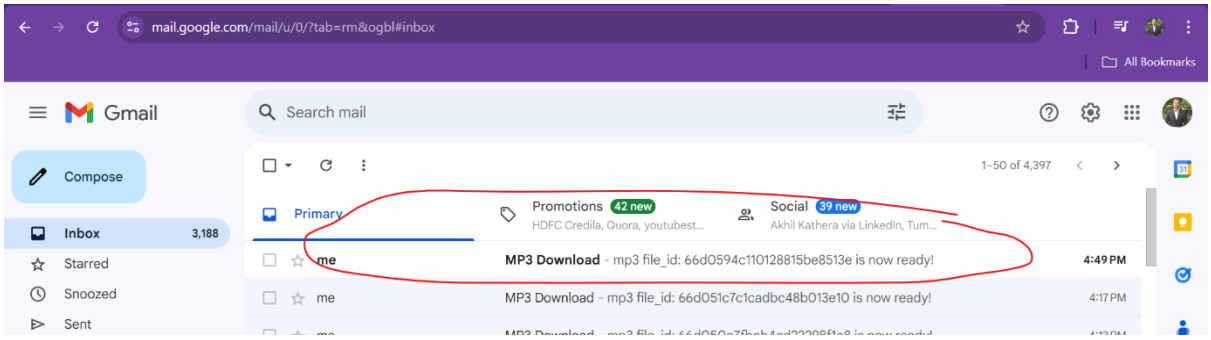
```
ubuntu@ip-10-5-1-157:~/microservices-python-app/assets$  curl -X POST -F 'file=@./video.mp4' -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6Ik
pXVCJ9.eyJ1c2VybmFtZSI6ImFobWVkaHNoYWlraDc4NkBnbWFpbC5jb20iLCJleHAiOjE3MjUwMTY3NDYsImlhdCI6MTcyNDkzMDM0NiwiYWRtaW4iOnRydWV9.hDIZlv36qo0y5ET8waCT47sNu
kRzNjDgUr8x4jJGeS0' http://13.235.76.100:30002/upload
success!ubuntu@ip-10-5-1-157:~/microservices-python-app/assets$ ^C
```

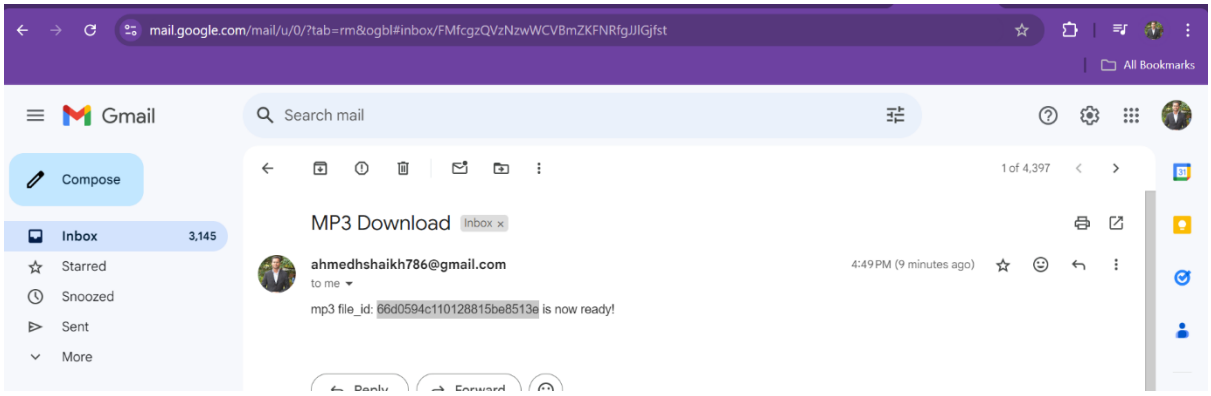
You can see that our video has been successfully uploaded.

Now to download that converted mp3 file we need to download it using the endpoint given below:

"curl --output video.mp3 -X GET -H 'Authorization: Bearer <JWT Token>' http://nodeIP:30002/download?fid=<Generated fid>"

But before running this command we will need the file id which we will get to our gmail, when we upload the video.





As you can have successfully downloaded the mp3 file:

```
ubuntu@ip-10-5-1-157:~/microservices-python-app/assets$  curl --output video.mp3 -X GET -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ
9.eyJlc2VybmFtZSI6ImFobWVkaHNoYWlraDc4NkBnbWFpbC5jb20iLCJleHAiOjE3MjUwMTY3NDYsImlhdCI6MTcyNDkzMDM0NiwiYWRtaW4iOnRydWV9.hDIZlv36qo0y5ET8waCT47sNukRzNj
DgUr8x4jJGeS0' "http://13.235.76.100:30002/download?fid=66d0594c110128815be8513e"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  288k    0  288k    0     0  2729k      0 --:--:-- --:--:-- --:--:-- 2750k
ubuntu@ip-10-5-1-157:~/microservices-python-app/assets$ ls
Rockstar.mp4  video.mp3  video.mp4
ubuntu@ip-10-5-1-157:~/microservices-python-app/assets$
```